MOHD SAQIB

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|SAQIB29ABUBAKAR - LEETCODE PROFILE | MOHD SAQIB | PORTFOLIO

Summary

M.Tech Computer Engineering (AI & ML) student at Jamia Millia Islamia with a B.Tech in Computer Engineering and a **BS in Data Science & Applications from IIT Madras**. Experienced in building **full-stack applications, LLM-based automation agents, RAG systems**, computer vision solutions, and **production-oriented AI applications**. **GATE 2025 & 2026** qualified in Computer Science and Data Science & AI.

Education

Faculty of Engineering and Technology, Jamia Millia Islamia, New Delhi, India
M.Tech in Computer Engineering (AI & ML) — **July 2026 – Present**
B-Tech in Computer Engineering | CGPA: **8.39/10 (First Division)** — **Sep 2022 - June 2026**

Indian Institute of Technology, Madras, India — **Jan 2024 - Dec 2026**
- BS in Data Science and Applications(Online) | CGPA: **7.24**

Rajkiya Pratibha Vikas Vidyalaya Lajpat Nagar School, New Delhi, India — **2020 - 2021**
- CBSE (Class XII with PCM), **Percentage: 94 %**

Govt Boys Senior Secondary School Jangpura, New Delhi, India — **2018 - 2019**
- CBSE (Class X), **Percentage: 93.4%**

Skills

Languages:
English, Hindi, Urdu, Arabic

Programming Languages:
C++, C, C#, Python, R, JavaScript, Java, SQL, Kotlin

Web Development:
HTML, CSS, Bootstrap, JavaScript, React.js, Node.js, Express.js, Jinja2, Flask, Tailwind CSS,Streamlit

Mobile Development:
Android SDK, Jetpack Compose, XML Layouts, Retrofit, Coroutines, Room DB, Firebase

Databases & Data: PostgreSQL, MySQL, MongoDB, SQLite, ChromaDB, FAISS

Data Science & Machine Learning:
NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, TensorFlow, XGBoost,PyTorch, Jupyter Notebook

Cloud & DevOps:
AWS, Google Cloud Platform (GCP), Docker, Kubernetes, Git, GitHub

Tools & IDEs:
VS Code,Android Studio, Postman, Excel, PowerPoint, Redis, Celery, Chart.js, ReportLab

Special Interests:
Quantum Computing, Automation Agents, LLM Integration, Prompt Engineering

Soft Skills:
Problem Solving, Data Structures & Algorithms (DSA), Team Collaboration, Time Management, Critical Thinking, Leadership, Emotional Intelligence

Generative AI & LLMs: LangChain, LangGraph, RAG, Prompt Engineering, LLM Integration, Tool Calling, Agentic AI, Multi-Agent Systems, Hugging Face, Pydantic, MCP

Work Experience

Instructor & Content Creator | *Udemy* — **Jan 2026 - March 2026**

- Designed and developed technical courses in **Python, Machine Learning, Generative AI, Agentic AI, and RAG**, including the **Complete Generative AI, Agentic AI & RAG Bootcamp**.
- Created and uploaded **30+ hours of structured video content** covering Python, NLP, Transformers, LLMs, Prompt Engineering, LangChain, RAG, Agentic AI, LangGraph, Multi-Agent Systems, MCP, Fine-Tuning, and AI application deployment.

- Architected hands-on curricula with **coding demonstrations, practical projects, quizzes, coding exercises, assignments, and real-world AI applications**.
- Previously developed and published courses including **Complete Machine Learning for Absolute Beginners, Python for Data Science, and Python for Absolute Beginners**.
- Built supporting course resources, project code, lecture notes, and practical AI application demonstrations to provide an end-to-end learning experience

Agnirva.com Space Community | Internship **June 2024 - July 2024**

- Gained comprehensive knowledge in satellite communication, spacecraft systems
- Demonstrated strong potential and capacity to apply space technology knowledge in the future.

Projects

Market Research & Brand Health Analysis – Cooking Oil Category **Aug 2026**

- Analyzed **106 respondents across 2 states and 3 age groups** to evaluate brand awareness, usage, satisfaction, continuation intent and recommendation across **11 sunflower-oil brands**.
- Conducted **TOM, Total Awareness and MOUB analysis**, identifying **Freedom as the strongest brand and Gold Drop as the second strongest** based on consistent awareness and usage leadership.
- Performed **Excel-based Welch's t-test, one-way ANOVA, paired t-test and Pearson correlation** to assess satisfaction differences and relationships among customer experience measures.
- Derived actionable insights from **NPS (-38.46)** and continuation intent (**5.38/10**), recommending stronger focus on **customer satisfaction, retention and advocacy**.
- **Skills:** Market Research, Excel, Statistical Analysis, Hypothesis Testing, ANOVA, t-tests, Correlation, NPS, Data Interpretation, Business Insights

Preparing for Tomorrow's Careers: Future Skills Required for Success in the AI Era **May 2026 – Aug 2026**

- Conducted a qualitative research study under IIT Madras BS (Strategies for Professional Growth) on future workforce skills required for success in the AI era.
- Collaborated in a 5-member interdisciplinary team to design the research methodology, analyze findings, and develop recommendations based on **6 industry expert interviews** and global reports from the **WEF Future of Jobs Report 2025, McKinsey, Harvard Business Review, and MIT Sloan**.
- Led project coordination, conducted a primary interview with a Software Engineer at TCS, performed thematic analysis of interview data, and synthesized insights on AI adoption, technology literacy, analytical thinking, lifelong learning, engineering management, and career planning.
- Developed professional project reports, case studies, influence models, and delivered a live team presentation summarizing industry insights and future-ready career recommendations.
- **Skills:** Qualitative Research, Interviewing, Thematic Analysis, Team Leadership, Project Management, Communication, Literature Review, AI & Future of Work, Career Planning, Presentation Design

Vision-Based Queue Analytics: Real-Time Queue Length & Waiting Time Estimation ***Jan 2026 – May 2026***

- Developed a near **real-time computer vision** system for **automated queue monitoring using YOLOv8s**, multi-object tracking, and ROI-based spatial analysis on CCTV video streams.
- Fine-tuned YOLOv8s on the **CrowdHuman dataset** (6k+ training images), achieving 93.3% mAP@0.5 for robust person detection under crowd occlusion and varying viewpoints.
- Designed a **queue-ordering pipeline using polygon ROI filtering**, geometric projection, and adaptive **EMA-based temporal modeling** to estimate queue length and waiting time dynamically.
- Implemented **persistent identity tracking and timestamp-based analytics** for service-time estimation, **achieving ~65% waiting-time prediction accuracy** and up to 78% consistency in stable queue conditions.
- Built a **real-time analytics dashboard** using Streamlit/Flask for monitoring queue statistics, enabling scalable deployment for hospitals, airports, retail stores, and smart-city environments.
- **Tech Stack**: Python, YOLOv8, OpenCV, ByteTrack, PyTorch, Streamlit, Flask, NumPy, Pandas

EventFlow: Sound & Light Rental Operations Platform ***Feb 2026 – April 2026***

- **Architecture & Scale:** Architected a scalable, type-safe monorepo platform using **Next.js 16, React 19, and Bun**. Managed 18+ PostgreSQL tables via **Drizzle ORM** and 12+ **oRPC** API routers for complex rental workflows.
- **Security & Multi-Tenancy:** Engineered a strict, organization-scoped **Role-Based Access Control (RBAC)** system using Better Auth (Owner, Event Head, Staff), successfully preventing cross-tenant data leaks and securing financial endpoints.
- **Advanced Feature Integration:** Developed a natural language **AI Operations Assistant** (Ctrl+K) for rapid command execution, alongside automated PDF/Excel invoice generation and **Gmail SMTP** integration for seamless client billing.
- **Quality Assurance & Testing:** Enforced strict API data validation using **Zod** and achieved a **100% pass rate across 114 Vitest integration tests**, ensuring zero runtime API errors and robust business logic.
- **Agile Leadership:** Led backend engineering and peer code reviews for a 4-member Agile team, utilizing Jira/Kanban workflows and GitHub PRs to deliver all 5 project milestones on strict academic deadlines.

Visual Question Answering (VQA) System – Image & Video (Minor Project) **Nov'25**

- Designed and implemented a unified **deep learning system** that answers **natural-language questions** from both **images and videos** using **multimodal AI**.
- Built an **end-to-end pipeline using BLIP (ViT + BERT**) for visual–text fusion, enabling accurate reasoning over visual content.
- Implemented a **lightweight frame-based Video VQA approach** using **OpenCV** to extract keyframes and aggregate predictions via confidence-weighted scoring.
- Developed a **web-based interactive interface with Flask** that supports **image/video upload**, question input, and real-time AI responses.
- Evaluated the system on the VQAv2 dataset, achieving **~62% accuracy**, and analyzed challenges such as hallucination and lack of temporal reasoning.
- **Tech Stack:** Python, PyTorch, HuggingFace Transformers, BLIP, OpenCV, Flask, HTML, CSS, JavaScript.

Smart Hospital Management System **Sept'25**

- Developed a **full-stack hospital management platform** to streamline patient registration, doctor scheduling, appointments, and medical records.
- Implemented **multi-role workflows for Admin, Doctors, Receptionists, and Patients** with secure authentication and role-based access control.
- Enabled digital patient history, **appointment tracking, and automated report generation**, reducing manual paperwork and improving operational efficiency.
- Built responsive dashboards for real-time monitoring of beds, appointments, and doctor availability.
- Tech Stack: Flask / Node.js, MySQL / SQLite, HTML, CSS, JavaScript, Bootstrap, Jinja2, Chart.js, ReportLab.

Quiz Master V2 – Multi-User Exam Portal **June'25**

- Developed a full-stack multi-user web app for quiz management using **Flask (API), VueJS (UI), SQLite, Redis, Celery,** and **Bootstrap**, supporting both admin and user roles.
- Implemented complete role-based access control: admin **creates subjects, chapters, quizzes, and MCQs**, while users register, attempt timed quizzes, view history, and export results.
- Integrated **async background jobs using Redis + Celery** for:
    - **Daily Google Chat/email reminders to users** about new quizzes
    - **Monthly user activity reports** via email.
    - **CSV export**s of quiz history (for both admin and users)
- Added performance optimization via **caching, summary charts, and responsive UI**, ensuring smooth functionality across devices and platforms.

ML Regression Project – Kaggle Competition **May'25**

- Participated individually in a **Kaggle-hosted regression competition** as part of the Machine Learning Project course; built and evaluated multiple ML models (Linear Regression, SVM, KNN, MLP, Ensembles) to improve prediction

accuracy.

- Performed **end-to-end ML pipeline**: data loading, EDA, preprocessing, model training, hyperparameter tuning, and cross-validation, **using scikit-learn, XGBoost, pandas, seaborn, matplotlib**.

- Achieved a final submission score **above 0.45 cutoff**, meeting all milestones and eligibility criteria for viva; demonstrated understanding of model evaluation, overfitting handling, and feature engineering.

- Successfully defended **model approach in Level 1 & 2 viva exams**, showcasing an understanding of algorithm internals, mathematical foundations, and practical deployment considerations.

Business Optimization for Local Electronics Retailer **Mar'25**

- Collected and **digitized 30+ days of real-time sales** and **inventory data from a local shop**, enabling structured analysis using Excel, Google Sheets, and Python (pandas, matplotlib, seaborn).

- Performed **sales, inventory, and profit margin analysis through visual tools like heatmaps, turnover charts, and trend lines** to identify **inefficiencies and price-sensitive consumer** behavior.

- **Proposed actionable strategies including digital inventory tracking, pricing optimization, and UPI-based discounting,** tailored to the shop's operations.

- Successfully improved sales and digital payment usage; the shop adopted structured data practices, resulting in better inventory planning and improved profit margins.

LLM-based Automation Agent **Jan'25**

- Built a robust **automation agent using GPT-4o-mini** that accepts plain-English tasks and executes them via an API (`/run?task=...`), returning verifiable results.

- Implemented **10 real-world Ops tasks** (file formatting, data parsing, log extraction, JSON sorting, SQL querying, markdown indexing) using shell tools, Python, and LLMs as task interpreters.

- Extended support for business automation including Git commits, API fetches, image compression, and CSV filtering—while enforcing security constraints like no data deletion or external access.

- Dockerized and deployed the application using AI Proxy Token integration and verified via automated testing scripts

TDS Solver – LLM API for Assignment Questions **Feb'25**

- Developed and deployed a **production-ready REST API** (`/api/`) that answers **assignment questions from 5 graded sets**, using **LLM-based inference** (via GPT-4o-mini).

- Supported **file uploads (e.g., `.csv`, `.zip`)** via `multipart/form-data` and used programmatic parsing and extraction to return **precise answers** in JSON format.

- Deployed the app on **Vercel** with public access and shared **GitHub repo (MIT license)** meeting all submission criteria (testable endpoint + clean API response).

- Enabled fast, scalable model-backed answers suitable for academic automation and passed evaluation against randomized test questions.
- (`validate.py`, `run.sh`) for 100% task accuracy.

Canteen Management System **May'24**

- Designed and developed a scalable system to manage **real-time orders and inventory** across **4 parallel canteens**, optimizing service speed and operational control.

- Enabled **multi-canteen handling**, dynamic menu updates, and livestock tracking with role-based access for admins and users.

- Integrated a clean, responsive UI with **Bootstrap**, ensuring a seamless user experience for order placement and inventory overview.

- **Tech Stack**: Node.js, Express.js, MySQL, HTML, CSS, JavaScript, Bootstrap.

Smart Home Services Platform **Sep'24**

- Developed a **multi-role web applicatio**n to connect customers with home service professionals (e.g., painting, cleaning, Wi-Fi setup), with dedicated dashboards for **Admin, Users, and Service Providers**.

- Implemented core features, including **user authentication (Flask-Login)**, **service booking, real-time dashboards, and PDF report generation**, using a well-structured **SQLite database** and foreign key relationships.

- Integrated **search, models,** and **analytics** with **JavaScript, Chart.js,** and added **dynamic HTML rendering** via **Jinja2 templates** and **Bootstrap for responsive design.**

- **Tech Stack**: Flask, SQLite, Flask-Login, Jinja2, Bootstrap, JavaScript, ReportLab, Chart.js.

Academic and Extracurricular Achievements

- **GATE Qualified (2025 & 2026):**
  - **2026 (CS & IT): AIR 5633 among 2.11 lakh candidates.**
  - **2026 (Data Science & AI): AIR 4412 among 69,000+ candidates.**
  - **2025 (CS & IT): AIR 19312 among 1.7 lakh candidates.**
- **JEE Advanced(2022)** qualified: **CRL - 24106** among 1.6 lakh students
- **JEE Main(2022**): **CRL -13654** among 10 lakh students
- Ranked among **the top 20 students** in the Computer Engineering department at Jamia Millia Islamia
- Maintained exceptional academic consistency throughout schooling—ranked **1st–2nd (Grades 1–5)**, **1st (Grades 6–10)**, and **Top 3 (Grades 11–12)**.
- **Completed memorization** of the **Holy Qur'an (**Demonstrated high-retention memory and long-term disciplined commitment)
- **Zad Academy (Islamic Studies):** Pursuing a 2-year academic program (2026-27) in 7 core subjects including Fiqh and Arabic; dedicating 15 hours weekly alongside dual-degree engineering studies.
- Maintained a **300+ day continuous coding streak** on LeetCode, solving **500+** problems.

Relevant Coursework

- **Core Computer Science**:
  Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, System Software, Compiler Design, Database Management Systems, Microprocessors, Software Engineering, Software Testing, Computer Organization, Computer Architecture, Parallel and Distributed Computing, Mobile Computing, and IOT.
- **Programming & Tools**:
  Programming in Python, Programming Concepts using Java, Programming, Data Structures & Algorithms in Python, Tools in Data Science, Linux Fundamentals
- **Data Science & AI/ML**:
  Machine Learning Foundations, Machine Learning Techniques, Business Data Management, Statistics for Data Science I & II, Mathematics for Data Science I & II, AI: Search Methods for Problem Solving, Data Mining, Natural Language Processing, Computer Vision, and Digital Image Processing, Introduction to Large Language Models, Deep Learning

- **Mathematics & Theoretical Foundations**:
  Discrete Mathematics, Engineering Mathematics I–IV, Automata Theory, Fundamentals of Computing, Computational Thinking, Quantum Algorithms and Cryptography, Blockchain and its Applications
- **Soft Skills & Cognitive Science:**:
  Communication Skills, Emotional Intelligence, Introduction to Psychology